A Fixed Aperiodic Greedy Pair-Sum-Avoiding Sequence

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Abstract

With the help of GPT-5.6 Sol, we construct an explicit finite set of positive integers whose greedy pair-sum-avoiding extension has a non-eventually periodic sequence of consecutive gaps. The construction consists of a nonperiodic scale-eight sumset controller embedded in three residue classes modulo 49, together with a finite modular shield. For the fixed 21-element seed

$$A = \{1, 2, 3, 5, 7, 13, 22, 27, 28, 32, 36, 40, 47, 48, 52,\\ 63, 71, 77, 81, 89, 97\},$$

we give an explicit infinite set S, prove $t \in S \iff t \notin S + S$ for every $t > 97$, and prove that the membership indicator of S is not ultimately periodic. The exact recurrence verifies the least-admissible-next-term rule, with equal summands allowed, and implies that the gap sequence of the greedy extension is not eventually periodic.

1 The greedy extension and the main theorem

Write $\mathbb{N} = \{1, 2, 3, \ldots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. For subsets $X, Y \subseteq \mathbb{N}_0$ and $c \in \mathbb{N}_0$, put

$$X + Y = \{x + y : x \in X,\ y \in Y\}, \qquad c + X = \{c + x : x \in X\}, \qquad X^{\mathsf{c}} = \mathbb{N}_0 \setminus X.$$

Equal summands are allowed in every sumset.

Let $A = \{a_1 < \cdots < a_k\} \subseteq \mathbb{N}$ be a nonempty finite set. Install all elements of A as the initial terms. Having chosen $a_1 < \cdots < a_n$, with $n \geq k$, define

$$a_{n+1} = \min\{t > a_n : t \notin A_n + A_n\}, \qquad A_n = \{a_1, \ldots, a_n\}. \tag{1}$$

The minimum exists because $A_n + A_n$ is finite. No admissibility condition is imposed on the initial set A. Let $d_m = a_{m+1} - a_m$ denote the consecutive gaps.

Theorem 1.1. *Let*

$$A = \{1, 2, 3, 5, 7, 13, 22, 27, 28, 32, 36, 40, 47, 48, 52,\\ 63, 71, 77, 81, 89, 97\}.$$

For the sequence generated from A by (1)*,*

$$\forall M \geq 1\ \forall p \geq 1\ \exists m \geq M \quad d_{m+p} \neq d_m.$$

Thus its consecutive gap sequence is not eventually periodic.

The following elementary lemma will identify an explicitly constructed set with the output of the least-next rule.

Lemma 1.2 (Exact recurrence implies the greedy rule)**.** *Let $S \subseteq \mathbb{N}$ be infinite, let $N \in S$, and put $A = S \cap [1, N]_{\mathbb{Z}}$. Suppose that*

$$t \in S \iff t \notin S + S \qquad (t > N). \tag{2}$$

Then the increasing enumeration of S is exactly the greedy extension of A.

Proof. Regard the least-next rule as scanning the integers $N+1, N+2, \dots$ in increasing order and selecting precisely the admissible ones. Before the first decision, the selected set is $A = S \cap [1, N]_{\mathbb{Z}}$. We prove by induction on $t > N$ that, after the decision at t, the selected set up to t is $S \cap [1, t]_{\mathbb{Z}}$. Before the decision at t, the induction hypothesis gives the selected set $S \cap [1, t-1]_{\mathbb{Z}}$. If $t = u + v$ with $u, v \in S$, then $u, v \geq 1$, and hence $u, v < t$. Consequently

$$t \in (S \cap [1, t-1]_{\mathbb{Z}}) + (S \cap [1, t-1]_{\mathbb{Z}}) \iff t \in S + S.$$

Thus t is admissible exactly when $t \notin S + S$, which by (2) is exactly when $t \in S$. This proves the induction. Equivalently, if $s_m < s_{m+1}$ are consecutive elements of S with $s_m \geq N$, then

$$s_{m+1} \notin S_m + S_m, \qquad s_m < t < s_{m+1} \Longrightarrow t \in S_m + S_m,$$

where $S_m = \{s_1, \dots, s_m\}$. Hence s_{m+1} is the least admissible integer larger than s_m, as required. □

2 A scale-eight sumset controller

For integers $a \leq b$, write $[a, b]_{\mathbb{Z}} = \{a, a+1, \dots, b\}$. We shall repeatedly use the elementary identity

$$[a, b]_{\mathbb{Z}} + [c, d]_{\mathbb{Z}} = [a + c, b + d]_{\mathbb{Z}} \qquad (a \leq b,\ c \leq d). \tag{3}$$

Indeed, only the reverse inclusion requires comment. If $a + c \leq z \leq b + d$, set $x = \max\{a, z - d\}$ and $y = z - x$. Then $a \leq x \leq b$, $c \leq y \leq d$, and $z = x + y$.

Define

$$L = \bigcup_{k \geq 0} [8^k, 4 \cdot 8^k - 1]_{\mathbb{Z}} \subseteq \mathbb{N}_0 \tag{4}$$

and

$$Y_0 = \{0\} \cup \bigcup_{k \geq 0} [4 \cdot 8^k, 8 \cdot 8^k - 1]_{\mathbb{Z}}, \tag{5}$$

$$Y_1 = [1, 3]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [16 \cdot 8^k - 1, 32 \cdot 8^k - 1]_{\mathbb{Z}}, \tag{6}$$

$$Y_2 = [0, 1]_{\mathbb{Z}} \cup [7, 15]_{\mathbb{Z}} \cup \bigcup_{k \geq 0} [64 \cdot 8^k - 1, 128 \cdot 8^k - 1]_{\mathbb{Z}}. \tag{7}$$

For every $k \geq 0$, the interval $[8^k, 8^{k+1} - 1]_{\mathbb{Z}}$ is the disjoint union

$$[8^k, 4 \cdot 8^k - 1]_{\mathbb{Z}} \,\dot\cup\, [4 \cdot 8^k, 8 \cdot 8^k - 1]_{\mathbb{Z}}.$$

These intervals partition $\mathbb{N}$. Together with $0 \in Y_0$, this gives

$$Y_0 = L^{\mathsf{c}}. \tag{8}$$

Lemma 2.1 (Controller identities). *The sets in* (5)–(7) *satisfy*

$$Y_1 = (Y_0 + Y_0)^{\mathsf{c}}, \tag{9}$$
$$Y_2 = (Y_1 + Y_1)^{\mathsf{c}}, \tag{10}$$
$$Y_0 = \{0\} \cup (1 + (Y_2 + Y_2)^{\mathsf{c}}). \tag{11}$$

Proof. We first compute the three self-sumsets. Put $I_k = [4 \cdot 8^k, 8 \cdot 8^k - 1]_{\mathbb{Z}}$. For $a = 8^k$,

$$0 + I_k = [4a, 8a - 1]_{\mathbb{Z}}, \qquad I_k + I_k = [8a, 16a - 2]_{\mathbb{Z}},$$

so these adjacent intervals fill $[4a, 16a - 2]_{\mathbb{Z}}$. If $i \le k$, then $8^i \le a$, and (3) gives

$$I_i + I_k = [4(8^i + a), 8(8^i + a) - 2]_{\mathbb{Z}} \subseteq [4a, 16a - 2]_{\mathbb{Z}}.$$

Every sum of two nonzero elements of Y_0 is of this form after interchanging the summands. Therefore

$$Y_0 + Y_0 = \{0\} \cup \bigcup_{k \ge 0} [4 \cdot 8^k, 16 \cdot 8^k - 2]_{\mathbb{Z}}. \tag{12}$$

Next put $C = [1, 3]_{\mathbb{Z}}$ and $J_k = [16 \cdot 8^k - 1, 32 \cdot 8^k - 1]_{\mathbb{Z}}$. For $a = 8^k$,

$$C + C = [2, 6]_{\mathbb{Z}},$$
$$C + J_k = [16a, 32a + 2]_{\mathbb{Z}},$$
$$J_k + J_k = [32a - 2, 64a - 2]_{\mathbb{Z}}.$$

The latter two intervals overlap and fill $[16a, 64a - 2]_{\mathbb{Z}}$. If $i \le k$, then

$$J_i + J_k = [16(8^i + a) - 2, 32(8^i + a) - 2]_{\mathbb{Z}} \subseteq [16a, 64a - 2]_{\mathbb{Z}},$$

because $8^i \le a$ and $8^i \ge 1$. Hence

$$Y_1 + Y_1 = [2, 6]_{\mathbb{Z}} \cup \bigcup_{k \ge 0} [16 \cdot 8^k, 64 \cdot 8^k - 2]_{\mathbb{Z}}. \tag{13}$$

Finally let $H = [0, 1]_{\mathbb{Z}} \cup [7, 15]_{\mathbb{Z}}$ and $K_k = [64 \cdot 8^k - 1, 128 \cdot 8^k - 1]_{\mathbb{Z}}$. Directly from (3),

$$H + H = [0, 2]_{\mathbb{Z}} \cup [7, 30]_{\mathbb{Z}},$$
$$([0, 1]_{\mathbb{Z}}) + K_k = [64a - 1, 128a]_{\mathbb{Z}},$$
$$([7, 15]_{\mathbb{Z}}) + K_k = [64a + 6, 128a + 14]_{\mathbb{Z}},$$
$$K_k + K_k = [128a - 2, 256a - 2]_{\mathbb{Z}},$$

where again $a = 8^k$. Since $64a + 6 \le 128a$, the two middle intervals overlap, and hence

$$H + K_k = [64a - 1, 128a + 14]_{\mathbb{Z}}.$$

This interval overlaps $K_k + K_k$, and together they fill $[64a - 1, 256a - 2]_{\mathbb{Z}}$. For $i \le k$, one also has

$$K_i + K_k = [64(8^i + a) - 2, 128(8^i + a) - 2]_{\mathbb{Z}} \subseteq [64a - 1, 256a - 2]_{\mathbb{Z}}.$$

It follows that

$$Y_2 + Y_2 = [0, 2]_{\mathbb{Z}} \cup [7, 30]_{\mathbb{Z}} \cup \bigcup_{k \ge 0} [64 \cdot 8^k - 1, 256 \cdot 8^k - 2]_{\mathbb{Z}}. \tag{14}$$

The complementary gaps between the long intervals in (12) are

$$[1,3]_{\mathbb{Z}}, \qquad [16\cdot 8^k-1, 32\cdot 8^k-1]_{\mathbb{Z}} \qquad (k\geq 0),$$

which proves (9). The complementary gaps in (13) are

$$[0,1]_{\mathbb{Z}}, \qquad [7,15]_{\mathbb{Z}}, \qquad [64\cdot 8^k-1, 128\cdot 8^k-1]_{\mathbb{Z}} \qquad (k\geq 0),$$

which proves (10).

To prove the feedback identity, subtract 1 from the blocks of L. The blocks of $L-1=\{\ell-1:\ell\in L\}$ corresponding to $k=0$ and $k=1$ are $[0,2]_{\mathbb{Z}}$ and $[7,30]_{\mathbb{Z}}$, respectively. For $k=j+2$, the corresponding block is

$$[8^{j+2}-1, 4\cdot 8^{j+2}-2]_{\mathbb{Z}} = [64\cdot 8^j-1, 256\cdot 8^j-2]_{\mathbb{Z}}.$$

Thus (14) is exactly

$$Y_2+Y_2=L-1. \tag{15}$$

For $q\geq 1$, equations (8) and (15) give

$$q\in Y_0 \iff q\notin L \iff q-1\notin L-1 \iff q-1\in (Y_2+Y_2)^{\mathsf{c}}.$$

Together with $0\in Y_0$, this is (11). $\square$

Lemma 2.2. *The membership indicator of Y_0 is not ultimately periodic.*

Proof. Suppose that there are $N\geq 0$ and $h\geq 1$ such that

$$n\in Y_0 \iff n+h\in Y_0 \qquad (n\geq N).$$

Choose $a=8^k>\max\{N,h\}$ and set $m=4a-h$. Then

$$a\leq m\leq 4a-1, \qquad m\geq N.$$

Hence $m\in L$, so $m\notin Y_0$, whereas $m+h=4a\in[4a,8a-1]_{\mathbb{Z}}\subseteq Y_0$. This contradicts the assumed periodicity. $\square$

3 A finite shield modulo 49

All sums in this section are taken in $\mathbb{Z}/49\mathbb{Z}$, with residues written as their representatives in $\{0,1,\dots,48\}$. Put

$$P=\{7,14,28\}, \qquad B=\{3,22,32,40,48\}, \qquad D=\{1,2,5,13,27,36,47\}, \tag{16}$$

and let $Q=B\cup P$.

Lemma 3.1 (Finite shield certificate)**.** *In $\mathbb{Z}/49\mathbb{Z}$,*

$$(B+B)\cup(P+B)\cup(D+B)=\mathbb{Z}/49\mathbb{Z}\setminus Q, \tag{17}$$

$$Q\cap\big((B+B)\cup(P+B)\cup(D+B)\cup(D+P)\big)=\varnothing, \tag{18}$$

$$(P+P)\cap Q=P. \tag{19}$$

Moreover, the only ordered pairs from $P\times P$ whose sum lies in Q are the three diagonal pairs

$$7+7=14, \qquad 14+14=28, \qquad 28+28=49+7. \tag{20}$$

Proof. The required finite calculation is displayed in full:

$$\begin{aligned}
B+B &= \{2,5,6,13,15,21,23,25,31,35,39,43,44,47\},\\
P+B &= \{1,5,6,10,11,13,17,19,27,29,31,36,39,46,47\},\\
D+B &= \{0,1,4,5,8,9,10,12,16,18,19,20,23,24,26,27,\\
&\qquad 30,33,34,35,37,38,39,41,42,45,46\},\\
D+P &= \{1,5,6,8,9,12,15,16,19,20,26,27,29,30,33,34,41,43\},\\
P+P &= \{7,14,21,28,35,42\}.
\end{aligned}$$

The union of the first three lines is

$$\mathbb{Z}/49\mathbb{Z} \setminus Q = \mathbb{Z}/49\mathbb{Z} \setminus \{3,7,14,22,28,32,40,48\}.$$

None of the first four lines meets Q. The last line proves (19), and direct inspection of the nine ordered pairs in $P \times P$ gives (20). □

We now define the infinite set

$$\begin{aligned}
S = D &\cup \{49q+b : q \in \mathbb{N}_0,\ b \in B\}\\
&\cup \{49q+7 : q \in Y_0\}\\
&\cup \{49q+14 : q \in Y_1\}\\
&\cup \{49q+28 : q \in Y_2\}.
\end{aligned} \tag{21}$$

Every integer in (21) is positive. Below 98, the five full B-rails contribute

$$\{3,22,32,40,48,52,71,81,89,97\}.$$

Since

$$Y_0 \cap \{0,1\} = \{0\}, \quad Y_1 \cap \{0,1\} = \{1\}, \quad Y_2 \cap \{0,1\} = \{0,1\},$$

the three controller rails contribute $\{7,28,63,77\}$. Together with the seven elements of D, this gives

$$\begin{aligned}
S \cap [1,97]_{\mathbb{Z}} = \{&1,2,3,5,7,13,22,27,28,32,36,40,47,48,52,\\
&63,71,77,81,89,97\}.
\end{aligned} \tag{22}$$

4 The exact greedy recurrence

Theorem 4.1. *For every $t > 97$,*

$$t \in S \iff t \notin S+S. \tag{23}$$

Proof. Write $t = 49q + r$, with $0 \le r < 49$. Since $t > 97$,

$$q \ge 2. \tag{24}$$

We first treat residues in $Q = B \cup P$. For $t > 97$, membership $t \in S$ can only come from one of the infinite rails in (21), and hence requires $r \in Q$. The six possible types of pair sums are

$$D+D, \quad D+B, \quad D+P, \quad B+B, \quad B+P, \quad P+P.$$

A $D + D$ sum is at most $94 < t$. By (18), none of the next four types can have residue in Q. By Lemma 3.1, a $P + P$ sum can return to Q only through one of the three diagonal pairs in (20).

It follows that, for the present $q \geq 2$, representation on the controller rails is governed exactly as follows:

$$\begin{aligned} 49q + 14 \in S + S &\iff q \in Y_0 + Y_0, \\ 49q + 28 \in S + S &\iff q \in Y_1 + Y_1, \\ 49q + 7 \in S + S &\iff q \geq 1 \text{ and } q - 1 \in Y_2 + Y_2. \end{aligned}$$

Indeed, the corresponding representations are

$$\begin{aligned} (49u + 7) + (49v + 7) &= 49(u + v) + 14, \\ (49u + 14) + (49v + 14) &= 49(u + v) + 28, \\ (49u + 28) + (49v + 28) &= 49(u + v + 1) + 7. \end{aligned}$$

The converse in each line follows from the exclusion of every other source type. Equations (9), (10), and (11) now show that a controller-rail integer is represented if and only if it is omitted from S. No integer $t > 97$ on a full B-rail is represented, by the preceding exhaustive six-type analysis. Thus (23) holds for every residue in Q. In particular, if an omitted controller integer is considered explicitly, the relevant complement identity supplies u, v in the source controller set, and the three displayed equations give a representation by elements of S.

It remains to treat $r \notin Q$. Such an integer is not in S: its residue excludes every infinite rail, and $t > 97$ excludes the finite set D. Equation (17) supplies a residue representation of one of the three types $B + B$, $D + B$, or $P + B$.

Suppose first that $r = b + b'$ in $\mathbb{Z}/49\mathbb{Z}$, with $b, b' \in B$. For the standard representatives, write $b + b' = 49c + r$, where $c \in \{0, 1\}$. Then

$$t = b + \big(49(q - c) + b'\big).$$

Both summands lie on full B-rails. The second is positive, since $q \geq 2$ and $c \leq 1$. If instead $r = d + b$ in $\mathbb{Z}/49\mathbb{Z}$, with $d \in D$ and $b \in B$, write $d + b = 49c + r$, where $c \in \{0, 1\}$. Then

$$t = d + \big(49(q - c) + b\big),$$

using the fixed selected element d and a full-rail element.

Finally suppose that $r = p + b$ in $\mathbb{Z}/49\mathbb{Z}$, with $p \in P$ and $b \in B$. The integers

$$e_7 = 7, \qquad e_{14} = 63 = 49 + 14, \qquad e_{28} = 28$$

belong to S, because $0 \in Y_0$, $1 \in Y_1$, and $0 \in Y_2$. Write $e_p = 49u_p + p$ and $p + b = 49c + r$. Here $u_p, c \in \{0, 1\}$, and

$$t = e_p + \big(49(q - u_p - c) + b\big).$$

By (24), the second quotient is nonnegative, so the second summand is a positive element of a full B-rail. This represents every integer whose residue lies outside Q.

All constructed summands are positive; since their sum is t, each is strictly smaller than t. We have therefore proved both directions of (23) for every $t > 97$. □

By (22), Theorem 4.1, and Lemma 1.2, the increasing enumeration of S is exactly the greedy extension of the seed in Theorem 1.1. All 21 seed elements are installed before the first greedy test, which occurs after 97. No condition is imposed retroactively; for example, the seed is not sum-free because $1 + 1 = 2$.

5 Nonperiodicity of the gaps

Lemma 5.1. *Let $T = \{t_1 < t_2 < \cdots\} \subseteq \mathbb{N}$ be infinite. If the consecutive gaps $t_{m+1} - t_m$ are eventually periodic, then the membership indicator of T is ultimately periodic.*

Proof. Suppose that the gaps have period $p \geq 1$ from index M onward. Let

$$h = \sum_{j=0}^{p-1} (t_{M+j+1} - t_{M+j}) > 0.$$

Every block of p consecutive gaps in the periodic tail has the same sum, so

$$t_{m+p} = t_m + h \qquad (m \geq M). \tag{25}$$

The infinite set $T \subseteq \mathbb{N}$ is unbounded. Hence, given an integer $n \geq t_M$, there is a unique $m \geq M$ such that $t_m \leq n < t_{m+1}$. By (25),

$$t_{m+p} \leq n + h < t_{m+p+1}, \qquad (n+h) - t_{m+p} = n - t_m.$$

Thus $n = t_m$ if and only if $n + h = t_{m+p}$. Equivalently,

$$n \in T \iff n + h \in T \qquad (n \geq t_M),$$

which is ultimate periodicity of the membership indicator. □

Proof of Theorem 1.1. The residue-7 rail of S satisfies the exact identity

$$49q + 7 \in S \quad \iff \quad q \in Y_0 \qquad (q \in \mathbb{N}_0). \tag{26}$$

Suppose, for contradiction, that membership in S were ultimately periodic with some period $h \geq 1$. Every positive multiple of an ultimate period is again an ultimate period, so $49h$ is an ultimate period of S. For every sufficiently large q, equations (26) and the $49h$-periodicity of S would therefore give

$$q \in Y_0 \iff 49q + 7 \in S \iff 49(q+h) + 7 \in S \iff q + h \in Y_0.$$

This contradicts Lemma 2.2. Hence the membership indicator of S is not ultimately periodic. By the contrapositive of Lemma 5.1, the consecutive gaps in the increasing enumeration of S are not eventually periodic. This proves the asserted quantified statement for every $M, p \geq 1$. □